

Introduction

This concise project writeup describes the motivation, design, implementation, and future scaling of my YiFaYuan (移法源) database. The database name is a play-on-words: it rhymes with “LiFaYuan,” the name for Taiwan’s legal parliament, but translates to “immigration-law-source” as opposed to “establish-law-court.” I welcome native Chinese speakers to tell me if this actually makes sense in practice.

Literature Review

Migration and asylum law is an interesting matter globally, but finds particular complexity in Taiwan. Like many countries, Taiwan is experiencing an ageing population requiring domestic care, as well as a growing industrial economy requiring consistent manual and factory labor. Additionally, as Ju (2026) highlights, young democracies like Taiwan have a soft power incentive to engage in “migration diplomacy” as a bid for greater international participation. Unique to Taiwan, though, is what leading scholar Kironska (2022) coins the “China Factor:” most people seeking asylum in Taiwan would likely be from Mainland China, which claims Taiwan as a province. Legislative action on asylum, where the target demographic is likely Chinese would-be asylees, therefore implies Taiwanese independence and is politically provocative. Unfortunately for prospective immigrants to Taiwan, this reality complicates any adjacent legislation, including the humdrum of visa and residence permits.

Therefore, it is well-worth thinking about immigration legislation as a structured process with complex and sometimes conflicting incentives. Taiwan needs immigrants, and this process necessitates legislation; but, political pressures constrain how, in what form, and by whom that legislation can come about.

While the Lifayuan hosts a well-maintained legislative portal and API, it is not immigration-specific and cannot be directly keyword-searched. The YiFaYuan database aims to reconfigure existing information on the OpenGovernment API more appropriate to an immigration studies use-case, and integrate it with other sources for additional research context.

Previous Directions & Explanation of Changes

I started this project interested in *foreign* migration bills enacted around the time that a country switches recognition to Taiwan to China (or vice-versa, in the case of Saint Lucia and Nauru). I quickly realized this was technically and conceptually infeasible: scraping different foreign websites, in different formats, in languages I don't speak, was not practical. Additionally, "bills" was not a clean-cut category with every country introducing legislation through different processes with different outputs. The sample size was smaller than expected, with additional gaps in data collection.

I then switched to using the OpenParliament API, hoping to merge bill information with diplomatic meetings. This was better, but impractical, as legislators are typically not present in diplomatic meetings. Name ambiguation with Chinese characters was also a challenge.

I am happy with the choice to focus on Parliament meetings rather than diplomatic, as it is cleaner with my data and lends itself to other analytical questions better-tailored to the legal process.

AI and Ethics Note

Artificial intelligence, specifically Claude, was frequently used for a variety of lower-level, non-conceptual coding tasks. This includes navigating the OpenParliament and GDELT APIs for keyword-based scraping, data cleaning and transformation including munging date troubleshooting, and most UI syntax. Schema design changes were requested, but considered and deployed on a case-by-case basis. Mistakes are my own.

As a migration-adjacent project, it is also worth thinking about database design in the context of IAFSM's ethical guidelines, one of which is specifically *competence*. While their focus is appropriate methodology and training, their point stands that "do no harm" relates to both malicious and negligent behavior. In a human research context, poor database design that leaves personal information susceptible to damage, exposure, or other abuse very much fits in the latter category. As such, thoughtful design and competent administration should be considered ethical imperatives, a framework I have tried to carry through this course.

Design

I was initially primarily interested in legislators, speeches, bill proposals, and contemporaneous news coverage of migration-related events. The former three are openly available on the Lifayuan's API (Chinese-only: <https://www.ly.gov.tw/Pages/List.aspx?nodeid=154>), which I scraped into CSV files with a keyword-based R script. I ran this script for both the speech API (keyword search applied to the "meeting content" domain), and the bill proposal API (keyword search applied to the "BillName" domain). The set of keywords I used are included in the following table:

Chinese [original]	English [translation]
移民, 入出境, 難民, 庇護, 居留, 無國籍, 驅逐, 遣返, 偷渡, 歸化, 移工	Immigration, entry/exit, refugee, asylum, residence, statelessness, deportation, repatriation, illegal immigration, naturalization, foreign worker

Regarding the event-based data, this was retrieved from GDELT via the bigQuery API in R, using the same keyword search as above.

Data

Both APIs returned structured data in spreadsheet form. I initially had three spreadsheets: bills proposed, meetings in which speeches were given, and news headlines. From the "cosignatory" and "speechers" domains of the Lifayuan data, which contained lists of speakers, I was able to transform the data into a new "legislators" table that matched each legislator with the speeches they'd given and bills they'd proposed or cosigned. These four CSVs represented the core domains I was interested in: legislators, bills, speeches, and events. The schema section below discusses the structural complications with these four entities, and why I ultimately had to decompose them for database integrity.

Schema

My core interests were legislators, bills, speeches, and events. When I set out to make a schema, I realized a few issues with this designation.

First, "bills" was a broad designation that failed to capture the realities of the legal process. Legislators rarely propose new bills out of the blue; most of the time, specific sections of existing bills are proposed for amendment, with each proposal requiring multiple legislative sessions. Therefore, "bills" had to become three separate hierarchical tables: main bill (e.g. the 移民法, immigration law), section of the bill up for amendment (e.g. "article 51 of the immigration law"), and then instance of legislation on that article (e.g. "the October 2016 hearing on the amendment of article 51 of the Immigration law"). This is admittedly an imperfect hierarchy, because a minority of proposals target multiple sections of a law. However, this works for a majority of use-cases and is the most theoretically defensible, especially as the relationships exist between legislative *action* and not overall bills.

Second, the relationship between "legislator" and "speeches" failed to capture the context of where those speeches took place, and also limited future expansion of general assembly meeting data. Speeches take place within meetings, where other items are discussed, and other legislators are in attendance. Additionally, tangential speeches may be given in the context of a main bill. To capture the potential link between bills and meetings, and explore meeting agendas in future expansions, it made more sense to link speeches to meetings rather than purely to legislators.

The amended schema is included below, as seen in PgAdmin:

Though the primary key is typically an ID, each name constraint still requires a text entry, since null values indicate a parsing error. Other constraints function to keep the data consistent, such as ‘role’ in the sponsorship table being restricted to ‘proposer’ or ‘consignatory’ to match the OpenParliament format.

In the case of the bill hierarchy, each lower-level designation is specified with “delete cascade” on it overarching designation. This is good practice with weak entities because amendments are meaningless without a reference; if a main bill is added or removed from the database, a floating “section 51” is confusing and tedious to remove.

Finally, the “date” constraint addresses ongoing problems with minguo versus YYYY date format. For consistency, this dataset only accepts Western-style dates as these are more compatible with data cleaning tools, and more interpretable to a broader audience.

Implementation

Since my data was initially imported as CSV files in R, it made the most sense to clean and transform them locally, then export them to PostGre. Since my priority was a clean mapping between theory and structural integrity, I first designed my schema in pgAdmin. It initially threw me some logical errors before running smoothly. Then, I transformed my data in R to match the tables represented in my schema. Conceptually, this was actually quite useful, since it gave me a clear view of exactly how data would be stored in my database. It also gave me the chance to manually check for errors, of which I found and addressed the following:

1. Date formats were mismatched. I changed minguo to YYYY, and dd/mm/YYYY to mm/dd/YYYY.
2. Legislator names were initially stored in each row as a comma-separated list, where my schema expected text. This required wide-pivoting my speech data, then extracting each speaker name so I had one name to one speech.
3. Many headlines were in non-English, non-Chinese languages, or had inactive links. I removed entries using a script other than Latin, and had AI go through each link and delete the inactive or broken ones.

Once my core CSV files exactly matched the expected format in PostGre, I used dbConnect in R to upload them. Looking at the populated database in pgAdmin let me check that my data matched the initial schema, and that my constraints were working correctly (i.e. that dates were still not showing up as minguo).

Once I was satisfied with both my schema and populated data, I used SQLite to reimport the database in R as a .sqlite file. This was convenient to store locally as I worked on UI, since pgAdmin drained my laptop batter and I wanted to avoid consistently re-exporting the database for purely cosmetic changes.

I did the UI in R, primarily prompting Claude to make specific functions and continually refine them according to my aesthetic preferences and use-case. When I was satisfied with the local render in RShiny, I added my shinyapps.io token and officially published it to the site. Conveniently, once an app.R file is linked to your Shiny account and project, you can just click “republish” to update the application.

UI Considerations

The app is designed for a Chinese-English bilingual audience. Personally, I find bilingual layout confusing (since translations are oftentimes indirect and create a disambiguation problem), so I toggled the settings to be fully in English, or fully in Chinese, depending on researcher preference.

Due to personal preference, I also tried to “nest” things as much as possible: one page directs to others with multiple tabs, instead of the R default of putting everything on top of itself on the same page. This initially created some logical problems, for instance, assigning events to legislators that were inherited from the single event they’d attended. Tinkering with the UI reduced these problems, to the extent I can discern from using it.

Upon writing this reflection, I realized that “nesting” everything is only convenient for exploratory analysis, as there is no way to access the legislator or referendum main pages from the innermost query. So, I added an additional “legislator search” panel. It only accepts Chinese characters, but fuzzy string matching with pinyin or romanization was a daunting task within pgAdmin, and deferring to characters prevents name ambiguity issues.

Research Demonstration

The following queries show potential research use-cases of this database. Note that pgAdmin allows more elaborate queries than the app itself, which was designed to be exploratory and user-friendly.

1. Which legislators are the most active on immigration [i.e. return the five legislators that have given the most speeches and bills related to immigration, combined].

Query

Query History

AI Assistant

```
1 SELECT
2   l.name,
3   COUNT(DISTINCT s.sponsorship_id) AS bills_signe
4   COUNT(DISTINCT sp.speech_id)     AS speeches_gi
5   COUNT(DISTINCT s.sponsorship_id) + COUNT(DISTIN
6 FROM legislators l
7 LEFT JOIN sponsorships s ON s.leg_id = l.leg_id
8 LEFT JOIN speeches sp    ON sp.leg_id = l.leg_id
9 GROUP BY l.leg_id, l.name
10 ORDER BY total_activity DESC
11 LIMIT 5;
```

Data Output

Messages

Notifications

Showing rows: 1 to 5

Page No: 1 of 1

	name text	bills_signed bigint	speeches_given bigint	totalActivity bigint
1	何欣純	30	21	51
2	孔文吉	30	18	48
3	張宏陸	17	30	47
4	吳琪銘	18	27	45
5	王美惠	24	19	43

Ho Hsin-chun

MLY

何欣純



Member of the Legislative Yuan

[Wikipedia page]

Query returns Ho Hsin-Chun (DPP) as the most “active” member on migration bills. It is hard to tell whether this is because she is active and influential in parliament, in general; but either way, someone with a clear say in how migration bills move through the system.

2. Which legislators (since 2020) have given speeches about immigration, but never actually signed off on a bill?

QueryQuery HistoryAI Assistant

1SELECT DISTINCT l.name

2FROM legislators l

3JOIN speeches spON sp.leg_id = l.leg_id

4JOIN meetings mON m.meeting_id = sp.meeting_id

5WHERE m.meeting_date >= '2020-01-01'

6AND l.leg_id NOT IN (

7SELECT DISTINCT leg_id FROM sponsorships

8)

9ORDER BY l.name;

Data OutputMessagesNotifications

Showing rows: 1 to 3

Page No: 1 of 1

	name	text
1	張啓楷	
2	陳琬惠	
3	麥玉珍	

陳琬惠

Bills Sponsored

Speeches Given

Show 10 entries

Search:

Meeting	Date	Status	Unit	Bills Mentioned
第10屆第8會期內政委員會第11次全體委員會	10/23/-4659	散會(12:36)	內政委員會	
立法院第10屆第8會期內政委員會第7次全體委員會	09/25/-4659	散會(12:17)	內政委員會	
立法院第10屆第8會期內政委員會第5次全體委員會	09/10/-4659	散會(12:51)	內政委員會	
立法院第10屆第7會期內政委員會第9次全體委員會	03/03/-4659	休息(4月12日上午9時繼續開會)	內政委員會	
立法院第10屆第7會期內政委員會第4次全體委員會	01/29/-4659	休息(3月9日上午9時繼續開會)	內政委員會	Immigration Act

Click a row for full details and related news

Carol [Wan-Hui] Chen (TPP) seems peripheral to immigration proceedings, but has never signed a bill on the issue.

3. Show me some bills that came about during a period of anti-immigrant sentiment (i.e. return the three legislative actions with the lowest averaged [non-null] event “tone” per GDELT).

QueryQuery HistoryAI Assistant

8COUNT(e.event_id)AS news_count

9FROM legislative_actions la

10JOIN bill_sections bsON bs.section_id = la.section_id

11JOIN bills bON b.bill_id = bs.bill_id

12JOIN action_events aeON ae.action_id = la.action_id

13JOIN external_events eON e.event_id = ae.event_id

14WHERE e.avg_tone IS NOT NULL

15GROUP BY la.action_id, b.law_en, b.law_zh, bs.section_en, la.action_date, la.year

16HAVING COUNT(e.event_id) > 0

17ORDER BY mean_tone ASC

18LIMIT 3;

Data OutputMessagesNotifications

Showing rows: 1 to 3

Page No: 1 of 1

law_en	law_zh	section_en	action_date	year	mean_tone
New Economic Immigration ...	新經濟移民法	New Law Draft	2019-03-05	2019	-5.650
New Economic Immigration ...	新經濟移民法	New Law Draft	2019-02-22	2019	-5.500
Immigration Act	入出國及移民...	Amendment to Article(s) 第三十...	2019-03-22	2019	-5.203

02/14/-4663

Migrants rally against violence - Taipei Times

<http://www.taipeitimes.com/News/taiwan/archives/2019/03/25/2003712143>

-5.36912751677852

02/15/-4663

Over 100 migrant workers stage flash mob in Taipei Main Station | Taiwan News | Mar. 25, 2019 12:49

<https://www.taiwannews.com.tw/en/news/3665453>

-4.15841584158416

02/08/-4663

OPINION: Taiwan Must Discuss Migrant Worker Safety After Fatal Factory Fires | Eye On Taiwan

<https://www.eyeon taiwan.com/opinion-taiwan-must-discuss-migrant-worker-safety-after-fatal-factory-fires>

-7.03517587939698

02/08/-4663

Four illegal migrants

<https://menafn.com/1098269746/Four-illegal-migrants-arrested-in-Chiayi-Taiwan>

-10.4477611940298

The three bills are all clustered around March 2019, when there were a spate of deportations, a

fatal factory accident, and subsequent migrant worker protests, per the related news.

Scaling & Future Work

There are some key differences between the app and database, which I implemented due to other research directions I am interested in pursuing.

The obvious missing field is legislator party affiliation. This information was not available through the OpenParliament API, and name disambiguation was a challenge with web-scraping integration. There are also hundreds of legislators over the years with varying levels of importance; if this is a field I wish to add, it will be easiest to do so directly in pgAdmin and for the most active and prolific legislators. This would enable me to add the table “party,” and directly query the database for partisan legislative decisions.

Another future research direction is international-level partnerships. In the app, I currently only displayed “headline” and “URL” under the external events table; however, the database contains all information extracted from GDELT, including actors involved, event category, and “tone” (a sentiment analysis-related metric applied to each headline).

Conclusion

This database serves its purpose as a general reference tool for English- or Chinese-speaking researchers interested in querying the parliamentary meeting process. It can process amorphous bill structure, handles name and date ambiguity cleanly, and is configured for easy extension across other legal domains.

Works Cited

IASFM [International Association for the Study of Forced Migration] (2018). "Code of ethics: Critical reflections on research ethics in situations of forced migration."

<https://iasfm.org/wp-content/uploads/2018/11/IASFM-Research-Code-of-Ethics-2018.pdf>

Ju, A. (2026). Migration Diplomacy as a Path to Taiwan's Political Participation. *Center for Migration Studies*. <https://doi.org/10.1177/01979183261442887>

Kironska, K. (2022). Taiwan's Road to an Asylum Law: Who, when, how, and why not yet? *Human Rights Review*, 23:2, 241-264. [10.1007/s12142-021-00644-y](https://doi.org/10.1007/s12142-021-00644-y)